

DINA USMAN

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SUMMARY

Computer Science Engineering graduate (B.Tech, KTU Kerala, CGPA 8.06) with hands-on experience building and deploying full-stack and AI-integrated applications. Developed production-grade systems featuring real-time communication (WebSockets), LLM-powered functionality (Groq/Llama 3.3), async task processing (Celery/Redis), JWT authentication, automated testing, and CI/CD pipelines. Actively building toward AI/GenAI engineering — with applied experience in LLM API integration, prompt engineering, and emerging expertise in RAG pipelines, vector databases, and LangChain. Strong Python backend foundation (Django, DRF, FastAPI) combined with full-stack capability (React) and solid DevOps practices (Docker, GitHub Actions).

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C

Backend: Django, Django REST Framework, Django Channels, FastAPI, REST APIs

Frontend: React.js, Vite, HTML5, CSS3, React-Leaflet

Databases: PostgreSQL, SQLite, Redis

Authentication & Security: JWT Authentication, Role-Based Access Control (RBAC), API Rate Limiting

Testing & Quality Assurance: Pytest, Django Test Framework, unit testing, Postman, Swagger/OpenAPI

DevOps & Deployment: Docker, Docker Compose, GitHub Actions, Render, Vercel, Gunicorn, Azure App Service

AI & GenAI Engineering: Groq API, Llama 3.3, LangChain (learning), ChromaDB / Vector Databases (learning), Prompt Engineering, LLM API Integration, Embeddings, RAG Pipelines (building)

Computer Science Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Collaboration: Git, GitHub, Agile Development, Code Reviews, Technical Documentation

PROJECTS

SwiftChat — Full-Stack Real-Time Messaging Platform

Django · DRF · PostgreSQL · Redis · Django Channels · WebSockets · JWT · React · Cloudinary · Docker · GitHub Actions · Vercel · Render

Live: swiftchat-frontend-beta.vercel.app **GitHub:** github.com/dinamain/swiftchat-secure-messaging-platform

- Built and deployed a production-grade real-time messaging platform with a React/Vite frontend (Vercel) and Django/Daphne ASGI backend (Render), serving live users across web and mobile browsers.
- Implemented real-time bidirectional communication via Django Channels and Redis — async WebSocket consumer handling message delivery, typing indicators, online presence, delivery receipts, and live reaction updates.
- Engineered JWT auth with token blacklisting and silent auto-refresh via Axios interceptors, preventing unexpected session expiry.
- Built full messaging feature set: reply, forward, emoji reactions, pinned messages, search, notifications, and file/image attachments via Cloudinary with MIME type and size validation; full group management (create, rename, add/remove members, transfer ownership) and avatar upload via a custom RawMediaCloudinaryStorage class.
- Designed normalized PostgreSQL schema (7 models, self-referential FKs for replies/forwards) and documented 20+ REST endpoints with drf-spectacular (Swagger/OpenAPI).
- Containerized with Docker Compose and established GitHub Actions CI/CD pipelines with automated testing and deployment to Render on every push.

Event Registration Platform — Full-Stack Web App

Django · DRF · React/Vite · JWT · PostgreSQL · Docker · Celery · Redis · GitHub Actions · Pytest · Groq LLM API

Live: event-registration-platform-rho.vercel.app **GitHub:** github.com/dinamain/event-registration-platform

- Built a full-stack event management and registration platform with email-based authentication, role-based access control, and a staff-only admin dashboard for event and registration management.
- Integrated Groq's Llama 3.3 API for AI-generated event descriptions with prompt engineering for consistent structured output; implemented Celery/Redis for async email notifications and background jobs.
- Added interactive event location maps via React-Leaflet and applied DRF throttling on auth and AI endpoints; enforced duplicate-registration prevention at API and database levels.
- Wrote automated Pytest tests covering authentication, permissions, registrations, and business logic; established GitHub Actions CI pipelines on every push.

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science and Engineering

Graduated: June 2026

College of Engineering Thalassery, KTU, Kerala | CGPA: 8.06 / 10

CERTIFICATIONS

- NPTEL — Programming, Data Structures & Algorithms Using Python (IIT Madras)
- ICFOS — Machine Learning through Python (Feb 2024)